

devices in this world has grown a lot, and is likely to grow at a startling pace. Nanoscale devices are no longer a fantasy, we can now actually visualise or even see prototypes of ambient devices that are part of the décor. Connectivity has also improved a lot in the last few years, although how the traffic generated by a burgeoning ubi-comp environment will be borne by the backbone networks is still a question mark.

There has been major technological progress in many other aspects of ubi-comp as well: Unique ID schemes to identify the innumerable devices on the IoT, device- and system-level intelligence to make use of the information transacted by these devices, flexible displays, intuitive user interfaces and more. On the usage front too, there have been quite a few implementations of late, although these have been only at local or enterprise scales. Overall, we are moving steadily towards ubi-comp. But, as always, challenges remain.

A chip (or a code) in everything

Ubi-comp seems to require a chip in everything. Are current IC technologies capable of meeting this requirement in terms of functionality, volume and price-points?

"I think the current IC technologies suffice to meet the volume and price-points. Tags are getting cheaper," says Chiaki Ishikawa, senior researcher and international liaison, YRP Ubiquitous Networking Laboratory, Tokyo, Japan. "Embedded computer chips have reasonable horsepower to do what they are asked to do most of the time. However, we may need to investigate more on the reduction of power consumption of these IC chips," he adds.

The power needs of these ambient embedded devices are kind of special. These need to last almost forever without requiring replacement or losing power. Therefore in a ubi-comp future, it is likely that these embedded devices will mostly use renewable energy derived from ambient sources, such as light, heat or motion, with local energy storage capabilities to account for the

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intermittent nature of the energy source.

IC chips such as radio-frequency identification (RFID) tags are not the only means of implementing the IoT. There will also be many printed tags such as quick response (QR) codes. QR codes are basically square-shaped, black dots printed on a white background, wherein the printed pattern represents some information. We already see QR codes printed with advertisements these days—usually representing a URL. You can capture these codes on your mobile phone and directly go to the advertiser's webpage. Ishikawa feels that the volume and usage of QR codes in ubi-comp will be much larger than IC-based ones.

Networking—technologies aplenty, challenges too!

Ubi-comp will require a networking architecture integrating technologies ranging from personal-area networking (PAN) right up to the wide-area communication networks. The PAN and LAN appear easier to manage, as we have a wide choice of interesting technologies at that level, including Wi-Fi, RFID, ZigBee, near-field communications (NFC) and Li-Fi.

The big worry, however, is the backbone network. Ishikawa notes that the increased traffic to the trunk network lines in the last one year due to increased use of smartphones itself is quite alarming—reiterating the need to strengthen the trunk networks to sustain the growth of the Internet.

Neeraj Arora, director-service provider vertical, Cisco IBSG, says, "Specifically from mobile network technology perspective, platforms today are often

constrained. Challenges vary based on consumer- vs enterprise-specific service. While many consumer-specific mobile cloud apps require a high throughput, many enterprise M2M services (telematics, fixed telemetry, AMI, security, vending/POS) do not need high throughputs to function well. This is because they are only sending small amounts of data—often intermittently or even on an exception-only basis. However, they may have other requirements including optimised activation rates (ability of mobile packet gateways to scale out PDP context activations), stateful IP session failovers and IPv6 support, and higher QoS, which may pose challenges for mobile networks. To summarise, there is a need to optimise the mobile packet gateways for specific consumer or enterprise ubi-comp service requirement."

A deeper look into the mobile communication challenges

There are many aspects that need to be looked into as we transition from the comparatively simple networks of today to a more complex ubi-comp world, network capacity being just one of them. Some other issues that need attention are the interfaces and protocol definitions for communication between significant number of heterogeneous elements and a fundamentally new network architecture requiring universal broadband access at the edge nodes.

"The existing mobile Internet network will fall short of ubi-comp requirements in the areas of local multipoint connectivity, type of traffic supported, bandwidth needed, fault management and recovery actions, and the spectrum required," asserts Dr Suresh Borkar, a trend-watcher, consultant and communications expert who teaches at the Electrical and Computer Engineering Department of the Illinois Institute of Technology.

Borkar, explains each challenge in detail: Grid architectural concepts will entail processing distributed through the constellation of locally-embedded processors that work in parallel, and dealing